

QUESTIONS BASED ON PANDAS SERIES

Questions

Solutions

Q.1-	Given the following Series1 <table><tr><td>A</td><td>100</td></tr><tr><td>B</td><td>200</td></tr><tr><td>C</td><td>300</td></tr><tr><td>D</td><td>400</td></tr><tr><td>E</td><td>500</td></tr></table> Write the command to create above Series and then double the value in series and store in another series named Series2	A	100	B	200	C	300	D	400	E	500	import pandas as pd Series1=pd.Series([100,200,300,400,500],index=['A','B','C','D','E']) Series2=Series1*2 print(Series1) print(Series2) OUTPUT A 100 B 200 C 300 D 400 E 500 dtype: int64 A 200 B 400 C 600 D 800 E 1000 dtype: int64
A	100											
B	200											
C	300											
D	400											
E	500											
Q.2-	State whether True or False a. A series object is size mutable. b. A Dataframe object is value mutable	a. A series object is size mutable. (False) b. A Dataframe object is value mutable (True)										
Q.3-	Consider a given Series , Series1: 200 700 201 700 202 700 203 700 204 700 Write a program in Python Pandas to create the series and display it.	import pandas as pd Series1=pd.Series(700,index=range(200,205)) print(Series1) OUTPUT 200 700 201 700 202 700 203 700 204 700 dtype: int64										
Q.4-	Consider the following Series object, s IP 95 Physics 89 Chemistry 92 Math 95 i. Write the Python syntax which will display only IP. ii. Write the Python syntax to increase marks of all subjects by 10.	import pandas as pd s=pd.Series([95,89,92,95],index=['IP','Physics','Chemistry','Math']) print(s.index[0]) s=s+10 print(s) ANSWER: i) series_object.index[index number] ii) series_object=series_object+10 OUTPUT: IP 95 Physics 89 Chemistry 92 Math 95 dtype: int64 IP 105 Physics 99 Chemistry 102 Math 105 dtype: int64										

Q.5-	Consider a given series : SQTR <table><tr><td>QTR1</td><td>50000</td></tr><tr><td>QTR2</td><td>65890</td></tr><tr><td>QTR3</td><td>56780</td></tr><tr><td>QTR4</td><td>89000</td></tr><tr><td>QTR5</td><td>77900</td></tr></table> Write a program in Python Pandas to create and display the series.	QTR1	50000	QTR2	65890	QTR3	56780	QTR4	89000	QTR5	77900	<pre>import pandas as pd val1=[50000,65890,56780,89000,77900] idx=['QTR1','QTR2','QTR3','QTR4','QTR5'] SQTR=pd.Series(val1,index=idx) print(SQTR)</pre> OUTPUT: <pre>QTR1 50000 QTR2 65890 QTR3 56780 QTR4 89000 QTR5 77900 dtype: int64</pre>
QTR1	50000											
QTR2	65890											
QTR3	56780											
QTR4	89000											
QTR5	77900											
Q.6-	What will be the output produced by the following programming statements 1 & 2? import pandas as pd S1=pd.Series(data=[31,41,51]) print(S1>40) -->Statement1 print(S1[S1>40]) -->Statement2	Statement1- <pre>0 False 1 True 2 True dtype: bool</pre> Statement2- <pre>1 41 2 51 dtype: int64</pre>										
Q.7-	Given two series S1 and S2 S1 A39 B41 C42 D44 S2 A10 B10 D10 F10 Find the output for following python pandas statements? a. S1[: 2]*100 b. S1 * S2 c. S2[: : -1]*10	<pre>import pandas as pd S1=pd.Series([39,41,42,44],index=['A','B','C','D']) S2=pd.Series(10,index=['A','B','D','F']) print(S1[: 2]*100) print(S1 * S2) print(S2[: : -1]*10)</pre> OUTPUT: <pre>A 3900 B 4100 dtype: int64 A 390.0 B 410.0 C NaN D 440.0 F NaN dtype: float64 F 100 D 100 B 100 A 100 dtype: int64</pre>										
Q.8-	Given the following Series S1 and S2: S1 A10 B20 C30 D40 S2 A5 B4 C6 D8 Write the command to find the multiplication of series S1 and S2	<pre>import pandas as pd S1=pd.Series([10,20,30,40],index=['A','B','C','D']) S2=pd.Series([5,4,6,8],index=['A','B','C','D']) print(S1*S2)</pre> OUTPUT: <pre>A 50 B 80 C 180 D 320 dtype: int64</pre>										
Q.9-	Consider a given Series , Subject: <table><tr><td>ENGLISH</td><td>75</td></tr><tr><td>HINDI</td><td>78</td></tr><tr><td>MATHS</td><td>82</td></tr><tr><td>SCIENCE</td><td>86</td></tr></table> Write a program in Python Pandas to create this series	ENGLISH	75	HINDI	78	MATHS	82	SCIENCE	86	<pre>import pandas as pd mrk=[75,78,82,86] idx=['ENGLISH','HINDI','MATHS','SCIENCE'] Subject=pd.Series(mrk,index=idx) print(Subject)</pre>		
ENGLISH	75											
HINDI	78											
MATHS	82											
SCIENCE	86											

		OUTPUT: ENGLISH 75 HINDI 78 MATHS 82 SCIENCE 86 dtype: int64								
Q.10-	Consider the following Series object, “company” and its profit in Crores <table><tr><td>TCS</td><td>350</td></tr><tr><td>Reliance</td><td>200</td></tr><tr><td>L&T</td><td>800</td></tr><tr><td>Wipro</td><td>150</td></tr></table> i. Write the command which will display the name of the company having profit>250. ii. Write the command to name the series as Profit.	TCS	350	Reliance	200	L&T	800	Wipro	150	<pre>import pandas as pd profit=[350,200,800,150] idx=['TCS','Reliance','L & T','Wipro'] company=pd.Series(profit,index=idx) print(company[company>250]) company.name="Profit" print(company)</pre> OUTPUT: TCS 350 L & T 800 dtype: int64 TCS 350 Reliance 200 L & T 800 Wipro 150 Name: Profit, dtype: int64
TCS	350									
Reliance	200									
L&T	800									
Wipro	150									
Q.11-	Consider two objects a and b. a is a list whereas b is a Series. Both have values 10,20,25,50. What will be the output of the following two statements considering that the above objects have been created already a. print(a*2) b. print(b*2) Justify your answer.	<pre>import pandas as pd a=[10,20,25,50] b=pd.Series([10,20,25,50]) print(a*2) print(b*2)</pre> OUTPUT: Option a) will produce [10, 20, 25, 50, 10, 20, 25, 50] Option b) will produce 0 20 1 40 2 50 3 100 dtype: int64 Justification: In Option a) list elements is repeated two times, because a list is replicated when multiplied by any number, it does not allowed vector operation. In Option b) Series allows vector operation, that is why each element of the series has been multiplied by 2.								

Q.12-	Given a Pandas series called Sample, the command which will display the last 3 rows is . a. <code>print(Sample.tail(3))</code> b. <code>print(Sample.Tail(3))</code> c. <code>print(Sample.tails(3))</code> d. <code>print(Sample.Tails(3))</code>	Correct Answer: a. <code>print(Sample.tail(3))</code>
Q.13-	What will be the output of the following code? <pre>import pandas as pd s = pd.Series(6,index=range(0,5)) print(s)</pre>	OUTPUT: <pre>0 6 1 6 2 6 3 6 4 6 dtype: int64</pre>
Q.14-	If series s1 is having following data, <pre>1 6 3 1 5 3 7 5 9 4 11 8 13 7 15 4 17 6 19 7 dtype: int64</pre> What would be the result of the command <code>print(s1[3:6])</code>?	<pre>import pandas as pd s1=pd.Series([6,1,3,5,4,8,7,4,6,7],index=range(1,20,2)) print(s1)</pre> <pre>1 6 3 1 5 3 7 5 9 4 11 8 13 7 15 4 17 6 19 7 dtype: int64</pre> OUTPUT: <pre>print(s1[3:6])</pre> <pre>7 5 9 4 11 8 dtype: int64</pre>
Q.15-	What will be the output of the following code? <pre>import pandas as pd import numpy as np s = pd.Series(np.arange(10,50,10)) print(s) print(s.ndim) print(s.shape) print(len(s))</pre>	OUTPUT: <pre>0 10 1 20 2 30 3 40 dtype: int32 1 (4,) 4</pre>
Q.16-	Write a program to create a Series having 10 random numbers in the range of 10 and 20	<pre>import pandas as pd import random lst=[] for x in range(10): num=random.randint(10,20) lst.append(num) s = pd.Series(lst) print(s)</pre>

		OUTPUT: <pre> 0 20 1 19 2 18 3 13 4 19 5 12 6 16 7 17 8 11 9 11 dtype: int64 </pre>
Q.17-	Consider the following Series 's'- <pre> 0 4.0 1 5.0 2 7.0 3 NaN 4 1.0 5 10.0 dtype: float64 </pre> (i) Write a Python code to add 1 to all the elements. (ii) Write a code to replace all NaN with 0.	<pre> import pandas as pd import numpy as np s = pd.Series([4,5,7,np.nan,1,10]) print(s) # add 1 to the series s=s+1 print(s) s=s.fillna(0) print(s) </pre> <pre> 0 4.0 1 5.0 2 7.0 3 NaN 4 1.0 5 10.0 dtype: float64 0 5.0 1 6.0 2 8.0 3 NaN 4 2.0 5 11.0 dtype: float64 0 5.0 1 6.0 2 8.0 3 0.0 4 2.0 5 11.0 dtype: float64 </pre>
Q.18-	Predict the output of the following code. <pre> import pandas as pd import numpy as np data = {'one':'a','two':'b','three':'c'} s=pd.Series(data) print(s) print(s.size) </pre>	OUTPUT: <pre> one a two b three c dtype: object 3 </pre>
Q.19-	Create a Series object S1 using a python sequence [2,4,6,8] and default indices.	<pre> import pandas as pd data = range(2,10,2) S1=pd.Series(data) print(S1) </pre> OUTPUT: <pre> 0 2 1 4 2 6 3 8 dtype: int64 </pre>
Q.20-	Write the output of the following code fragment. <pre> import pandas as pd s2=pd.Series(["i","am", "a","student"]) print(s2) </pre>	OUTPUT: <pre> 0 i 1 am 2 a 3 student dtype: object </pre>

QUESTIONS BASED ON PANDAS SERIES (Part-2)

Q.21-	Write the output of the following code fragment. import pandas as pd s1=pd.Series(200,index=range(2,13,2)) print(s1)	OUTPUT: 2 200 4 200 6 200 8 200 10 200 12 200 dtype: int64
Q.22-	Write the output of the following code fragment. import pandas as pd s1=pd.Series(range(2,11,2), index=[x for x in "abcde"]) print(s1)	OUTPUT: a 2 b 4 c 6 d 8 e 10 dtype: int64
Q.23-	Write the output of the following code fragment. import pandas as pd import numpy as np x=np.arange(10,15) s3=pd.Series(index=x, data=x*2) s4=pd.Series(x**2,x) print(s3) print(s4)	OUTPUT: 10 20 11 22 12 24 13 26 14 28 dtype: int32 10 100 11 121 12 144 13 169 14 196 dtype: int32
Q.24-	Sequences section and contribution store the section name ('A','B','C','D','E') and contribution (8900,8700,7800,6500,nil) for charity. Your school has decided to donate more contribution by each section, so donation has been doubled. Write code to create series object that stores the contribution amount as the values and section name as indexes with data type as float32.	import pandas as pd import numpy as np idx=['A','B','C','D','E'] contr=[8900,8700,7800,6500,None] s1=pd.Series(data=contr, index=idx,dtype=np.float32) print("Donation by each section : Rs.") print(s1) print("Donation after the amount is doubled: Rs.") print(s1*2) OUTPUT: Donation by each section : Rs. A 8900.0 B 8700.0 C 7800.0 D 6500.0 E NaN dtype: float32 Donation after the amount is doubled: Rs. A 17800.0 B 17400.0 C 15600.0 D 13000.0 E NaN dtype: float32
Q.25-	Write the output of the following code fragment. import pandas as pd import numpy as np val1=np.arange(5.25,50,10.25) ser1=pd.Series(val1,index=['a','b','a','a','b']) print(ser1) print(ser1['a']) print(ser1['b'])	OUTPUT: a 5.25 b 15.50 a 25.75 a 36.00 b 46.25 dtype: float64 a 5.25 a 25.75 a 36.00 dtype: float64 b 15.50 b 46.25 dtype: float64

Q.26-	Consider a series object s10 that stores the number of students in each section of class 12 as shown below. First two sections have been given task for selling tickets @ Rs.100/- per ticket as a part of social experiment. Write code to create the series and display how much section A and B have collected. <table><tr><td>A</td><td>39</td></tr><tr><td>B</td><td>31</td></tr><tr><td>C</td><td>32</td></tr><tr><td>D</td><td>34</td></tr><tr><td>E</td><td>35</td></tr></table>	A	39	B	31	C	32	D	34	E	35	<pre>import pandas as pd import numpy as np S10=pd.Series([39,31,32,34,35],index=['A','B','C','D','E'], dtype=np.float32) print("Amount collected by Section A and B (in Rs.)") print(S10.head(2)*100)</pre> OUTPUT: <pre>Amount collected by Section A and B (in Rs.) A 3900.0 B 3100.0 dtype: float32</pre>							
A	39																		
B	31																		
C	32																		
D	34																		
E	35																		
Q.27-	Consider the series s4 as given below <table><tr><td>0</td><td>2.50</td></tr><tr><td>1</td><td>17.45</td></tr><tr><td>2</td><td>20.25</td></tr><tr><td>3</td><td>87.25</td></tr><tr><td>4</td><td>33.76</td></tr></table> What will be the output after executing the following: S4[0]=1.75 S4[2:4]= -23.74 print(S4)	0	2.50	1	17.45	2	20.25	3	87.25	4	33.76	<pre>import pandas as pd S4=pd.Series([2.50,17.45,20.25,87.25,33.76]) S4[0]=1.75 S4[2:4]= -23.74 print(S4)</pre> OUTPUT: <pre>0 1.75 1 17.45 2 -23.74 3 -23.74 4 87.25 5 33.76 dtype: float64</pre>							
0	2.50																		
1	17.45																		
2	20.25																		
3	87.25																		
4	33.76																		
Q.28-	Consider the following code- <pre>import pandas s1=pandas.Series([2,3,4,5,6],index=['a','b','c','d','e']) s1[1:5:2]=345.6 s1[2:4]= -14.65 print(s1)</pre> What will be the output after executing the code.	OUTPUT: <pre>a 2.00 b 345.60 c -14.65 d -14.65 e 6.00 dtype: float64</pre>																	
Q.29-	Consider the Series object s12 that stores the contribution of each section, as shown below: <table><tr><td>A</td><td>6700</td></tr><tr><td>B</td><td>8000</td></tr><tr><td>C</td><td>5400</td></tr><tr><td>D</td><td>3400</td></tr></table> Write code to modify the amount of section 'A' as 8800 and for section 'C' and 'D' as 7700. Print the changed object.	A	6700	B	8000	C	5400	D	3400	<pre>import pandas as pd import numpy as np idx=['A','B','C','D','E'] contr=[6700,8000,5400,3400] s12=pd.Series(contr, idx) print(s12) s12['A']=8800 s12[['C','D']]=7700 #or s12.loc['C':'D']=7700 print(s12)</pre>	OUTPUT: <table><tr><td>A</td><td>6700</td></tr><tr><td>B</td><td>8000</td></tr><tr><td>C</td><td>5400</td></tr><tr><td>D</td><td>3400</td></tr></table> <pre>dtype: int64 A 8800 B 8000 C 7700 D 7700 dtype: int64</pre>	A	6700	B	8000	C	5400	D	3400
A	6700																		
B	8000																		
C	5400																		
D	3400																		
A	6700																		
B	8000																		
C	5400																		
D	3400																		
Q.30	A Series object trainingdata consists of 2000 rows of data. Write a program to print (i) First 100 rows of data (ii) Last 5 rows of data	<pre>import pandas as pd trainingdata=pd.Series(275.50,index=range(2000)) # i) print(trainingdata.head(100)) #ii) print(trainingdata.tail(5))</pre>																	

Q.31-	Consider the following Series s3- <pre>a 1.5 b 3.0 c 4.5 d 6.0 e 7.5</pre> Now create the above series and find the output of the following commands- i) print(s3+3) ii) print(s3*2) iii) print(s3>3.0) iv) print(s3[s3>3.0])	<pre>import pandas as pd s3=pd.Series([1.5,3.0,4.5,6.0,7.5],[x for x in 'abcde']) print(s3+3) print(s3*2) print(s3>3.0) print(s3[s3>3.0])</pre> OUTPUT: <pre>a 4.5 b 6.0 c 7.5 d 9.0 e 10.5 dtype: float64 a 3.0 b 6.0 c 9.0 d 12.0 e 15.0 dtype: float64 a False b False c True d True e True dtype: bool c 4.5 d 6.0 e 7.5 dtype: float64</pre>																																																												
Q.32-	Consider the Series object s12 that stores the contribution of each section, as shown below: <pre>A 6700 B 8000 C 5400 D 3400</pre> Write code to create the series and display those sections that made the contribution more than Rs. 5600/-	<pre>import pandas as pd idx=['A','B','C','D'] contr=[6700,8000,5400,3400] s12=pd.Series(contr, idx) print(s12[s12>5600])</pre> OUTPUT: <pre>A 6700 B 8000 dtype: int64</pre>																																																												
Q.33-	Number of students in class 11 and 12 in three streams('Science', 'Commerce' and 'Humanities') are stored in two series objects c11 and c12. write code to find total number of students in class 11 and 12 , stream wise.	<pre>import pandas as pd stream=['Science','Commerce','Humanities'] c11=pd.Series(data=[30,40,50],index=stream) c12=pd.Series(data=[37,44,45],index=stream) print("Total no. of students:") print(c11+c12) # Series arithmetic</pre> OUTPUT: <pre>Total no. of students: Science 67 Commerce 84 Humanities 95 dtype: int64</pre>																																																												
Q.34-	Consider the series s1 and s2 and s3- <pre>S1 S2 S3 0 10 0 5 a 3 1 20 1 10 b 6 2 30 2 15 c 9 3 40 3 20 d 10 4 50 4 25 e 11 5 30 6 35</pre> Now find the output of the following- i) print(S1+S2) ii) print(S1*S3) iii) print(S1-S2)	<pre>import pandas as pd S1=pd.Series(data=[10,20,30,40,50],index=range(5)) S2=pd.Series(data=range(5,36,5),index=range(7)) S3=pd.Series(data=[3,6,9,10,11],index=[x for x in 'abcde']) print(S1+S2) print(S1*S3) print(S1-S2)</pre> OUTPUT: <table><tr><td>0</td><td>15.0</td><td>0</td><td>NaN</td><td>0</td><td>5.0</td></tr><tr><td>1</td><td>30.0</td><td>1</td><td>NaN</td><td>1</td><td>10.0</td></tr><tr><td>2</td><td>45.0</td><td>2</td><td>NaN</td><td>2</td><td>15.0</td></tr><tr><td>3</td><td>60.0</td><td>3</td><td>NaN</td><td>3</td><td>20.0</td></tr><tr><td>4</td><td>75.0</td><td>4</td><td>NaN</td><td>4</td><td>25.0</td></tr><tr><td>5</td><td>NaN</td><td>a</td><td>NaN</td><td>5</td><td>NaN</td></tr><tr><td>6</td><td>NaN</td><td>b</td><td>NaN</td><td>6</td><td>NaN</td></tr><tr><td></td><td></td><td>c</td><td>NaN</td><td></td><td></td></tr><tr><td></td><td></td><td>d</td><td>NaN</td><td></td><td></td></tr><tr><td></td><td></td><td>e</td><td>NaN</td><td></td><td></td></tr></table> dtype: float64 dtype: float64 dtype: float64	0	15.0	0	NaN	0	5.0	1	30.0	1	NaN	1	10.0	2	45.0	2	NaN	2	15.0	3	60.0	3	NaN	3	20.0	4	75.0	4	NaN	4	25.0	5	NaN	a	NaN	5	NaN	6	NaN	b	NaN	6	NaN			c	NaN					d	NaN					e	NaN		
0	15.0	0	NaN	0	5.0																																																									
1	30.0	1	NaN	1	10.0																																																									
2	45.0	2	NaN	2	15.0																																																									
3	60.0	3	NaN	3	20.0																																																									
4	75.0	4	NaN	4	25.0																																																									
5	NaN	a	NaN	5	NaN																																																									
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		e	NaN																																																											

Q.35-	Consider the Series object s12 that stores the contribution of each section, as shown below: <pre>D 6700 A 8000 B 5400 C 3400</pre> i) Write code to create the series and display all its values sorted in descending order. ii) Write code to display all its indices sorted in ascending order.	<pre>import pandas as pd idx=['D','A','B','C'] contr=[6700,8000,5400,3400] s12=pd.Series(contr, idx) # i) s12.sort_values(ascending=False) print(s12) #ii) s12.sort_index() print(s12)</pre> OUTPUT: <pre>A 8000 D 6700 B 5400 C 3400 dtype: int64 A 8000 B 5400 C 3400 D 6700 dtype: int64</pre>
Q.36	Given a Series object shown below: <pre>A 6700 B 8000 C 5400 D 3400 dtype : int64</pre> Why is following code producing error while working on Series object s13? <pre>import pandas as pd s13.index=range(0,5) print(s13)</pre>	This code is producing error because, object s13 has 4 elements only and the code line: <pre>s13.index=range(0,5)</pre> is trying to assign 5 index values to its elements: as range(0,5) produces 0,1,2,3,4 we can change the indexes of the Series object only when the new counting of indexes are same as original.
Q.37	What will be the output of the following program: <pre>import pandas as pd first=[7,8,9] second=pd.Series(first) s1=pd.Series(data=first*2) s2=pd.Series(data=second*2) print("Series1:") print(s1) print("Series2:") print(s2)</pre>	OUTPUT: <pre>Series1: 0 7 1 8 2 9 3 7 4 8 5 9 dtype: int64 Series2: 0 14 1 16 2 18 dtype: int64</pre>
Q.38	What is the output of the following program: <pre>import pandas as pd import numpy as np data=np.array(['Mon','Tue','Wed','Thu','Fri','Sat','Sun']) s=pd.Series(data) print(s[:4]) print(s[-4:])</pre>	OUTPUT: <pre>0 Mon 1 Tue 2 Wed 3 Thu dtype: object 3 Thu 4 Fri 5 Sat 6 Sun dtype: object</pre>
Q.39	What is the output of the following program: <pre>import pandas as pd import numpy as np data=np.array(['Mon','Tue','Wed','Thu','Fri','Sat','Sun']) s=pd.Series(data, index=[101,102,103,104,105,106,107]) print(s[[103,105,107]])</pre>	OUTPUT: <pre>103 Wed 105 Fri 107 Sun dtype: object</pre>

Q.40	What will be the output of the following: import pandas as pd D={'a':10,'b':11,'c':12} S=pd.Series(D,index=['b','c','d','a']) print(S)	OUTPUT: b 11.0 c 12.0 d NaN a 10.0 dtype: float64
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